

Supplementary materials: Estimating Cuba's 2021–2026 population collapse

Yasset Pérez-Riverol

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These materials accompany the main preprint. Preferred public estimates remain Model D; Model E is provisional and is not used for headline claims [Pérez-Riverol, 2026b,a].

1 Life expectancy series

Official life expectancy at birth peaked in 2011–13 (78.5 years) and fell to 77.7 in 2018–20, before the worst of the present crisis (Figure 1). An independent 2021 estimate places it near 71 years [Albizu-Campos Espiñeira, 2023, Brønnum-Hansen and Albizu-Campos Espiñeira, 2023]. Life expectancy re-expresses the same mortality shock rather than providing independent corroboration; its magnitude (about 6.5 years in a single year in the independent path) is exceptional for peacetime and large even relative to COVID-era losses documented internationally [Aburto et al., 2022, Msemburi et al., 2023]. Translated to years of life lost, crude excess of ~153,000 deaths in 2020–2025 implies more than one million person-years lost, concentrated among the elderly.

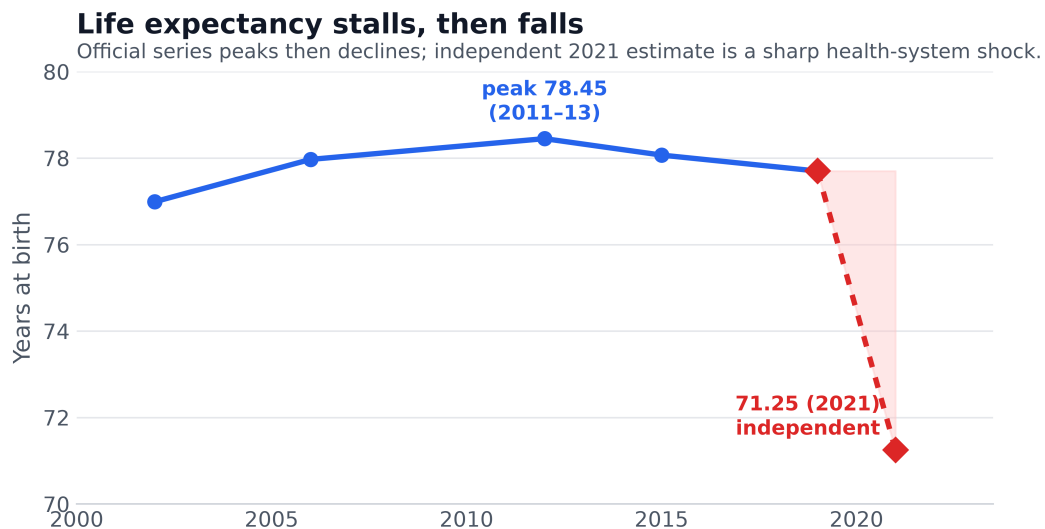


Figure 1: Life expectancy at birth: official triennial series and independent 2021 estimate.

2 Provincial intensity of exit

Emptying is national but unequal (Figure 2). Havana leads at net emigration of **–33 per 1,000** in 2025; western and central provinces empty faster than the east, yet no province loses less than about 2% of its people per year [ONEI, 2025].

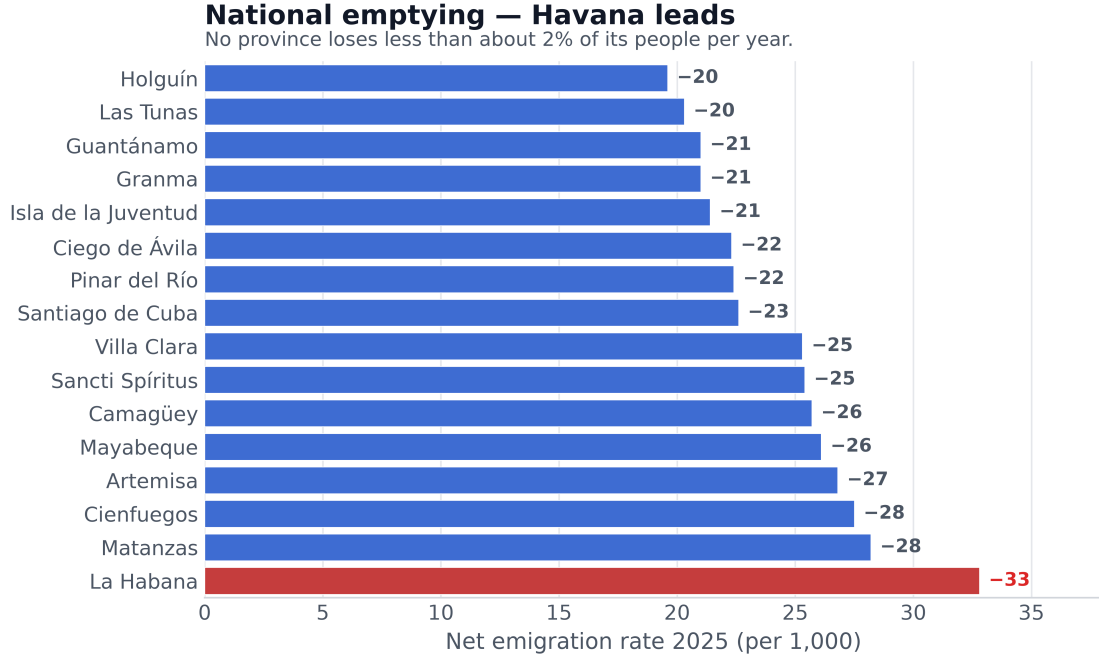


Figure 2: Net emigration rate by province, 2025.

3 Model E: vital reconstruction (provisional)

As a companion path, Model E reconstructs *true* deaths and births from a yearly health-system deterioration score S_t built from ageing, health-system decadence, and documented official underreporting (including COVID cause-attribution misuse as an integrity feature, not as a $\times N$ multiplier on all deaths). The approach follows the spirit of excess-mortality reconstruction under incomplete registration [Karlinsky and Kobak, 2021, Msemburi et al., 2023, Kishore et al., 2018], but uses a Cuban-specific deterioration score rather than a pandemic-only counterfactual.

Ageing enters an expected-death schedule; decadence raises an uplift $f(S_t)$; births stay near the register with crisis fertility and selective exit. Closing the identity $P^* = P_{\text{start}}^* + \sum (B^* - D^* - M^*)$ with a Model D-like migration bridge yields median end-2025 population about **8.96 million** (loss 2.13 million, 19.2% on the corrected base; migration share $\sim 88\%$). Figure 3 shows S_t and reconstructed versus registered vitals; Figure 4 places E beside A–D.

Preferred public claims remain Model D until an adversarial gate promotes E. Model E lands in the same loss band as D and raises the natural-decrease share only modestly ($\sim 12\%$ versus $\sim 9\%$ under D), so it does not yet displace D for public claims [Pérez-Riverol, 2026b].

Table 1: Models A–E medians (end-2025). Model E is provisional and reported only here.

Model	Main assumption	Loss (M)	%	Pop. (M)
A · conservative	ONEI-anchored	1.80	16.4	9.16
B · crisis-adjusted	under-reg. + independent migration	2.19	20.4	8.57
C · worst case	analog upper bound	2.41	22.6	8.26
D · sentinel (preferred)	IMR sentinel + triangulation	2.21	20.5	8.59
E · vital reconstr. (prov.)	ageing + HSDS $\rightarrow B^*, D^*$	2.13	19.2	8.96

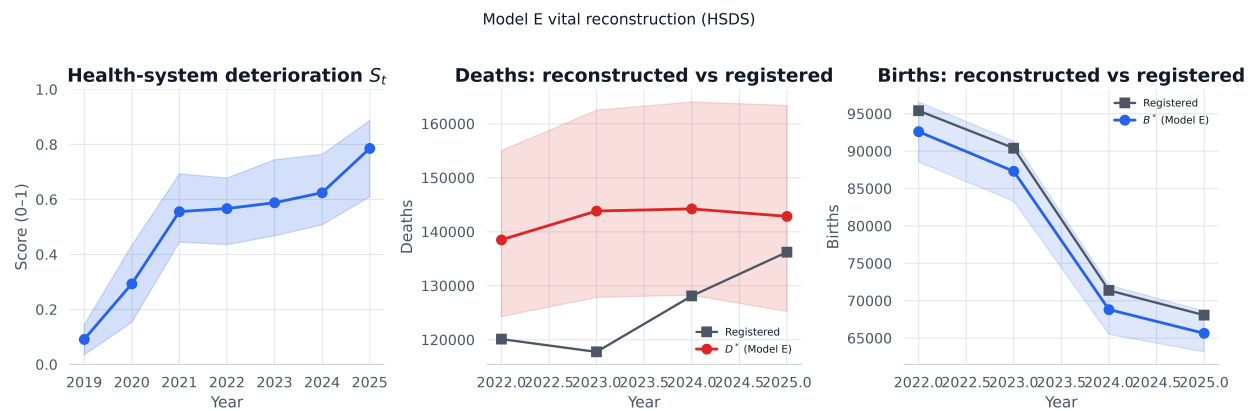


Figure 3: Health-system deterioration score S_t and Model E reconstructed deaths/births versus registered series (2022–2025).

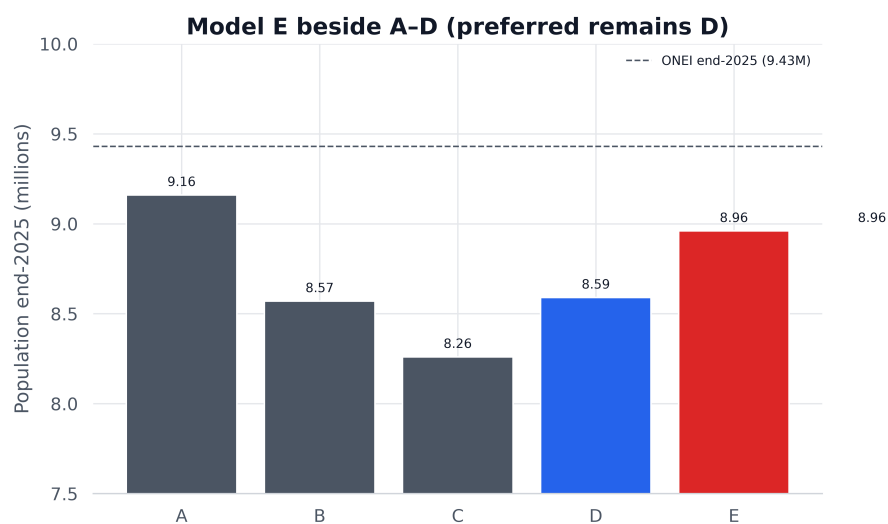


Figure 4: End-2025 population medians for Models A–E. Preferred public model remains D (blue); E (red) is provisional.

4 Data pointers

Scripts and JSON artifacts for HSDS / Model E live under `demographics/scripts/hsds_*.py` and `demographics/artifacts/` in the companion repository [Pérez-Riverol, 2026b].

References

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